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Data Snapshot

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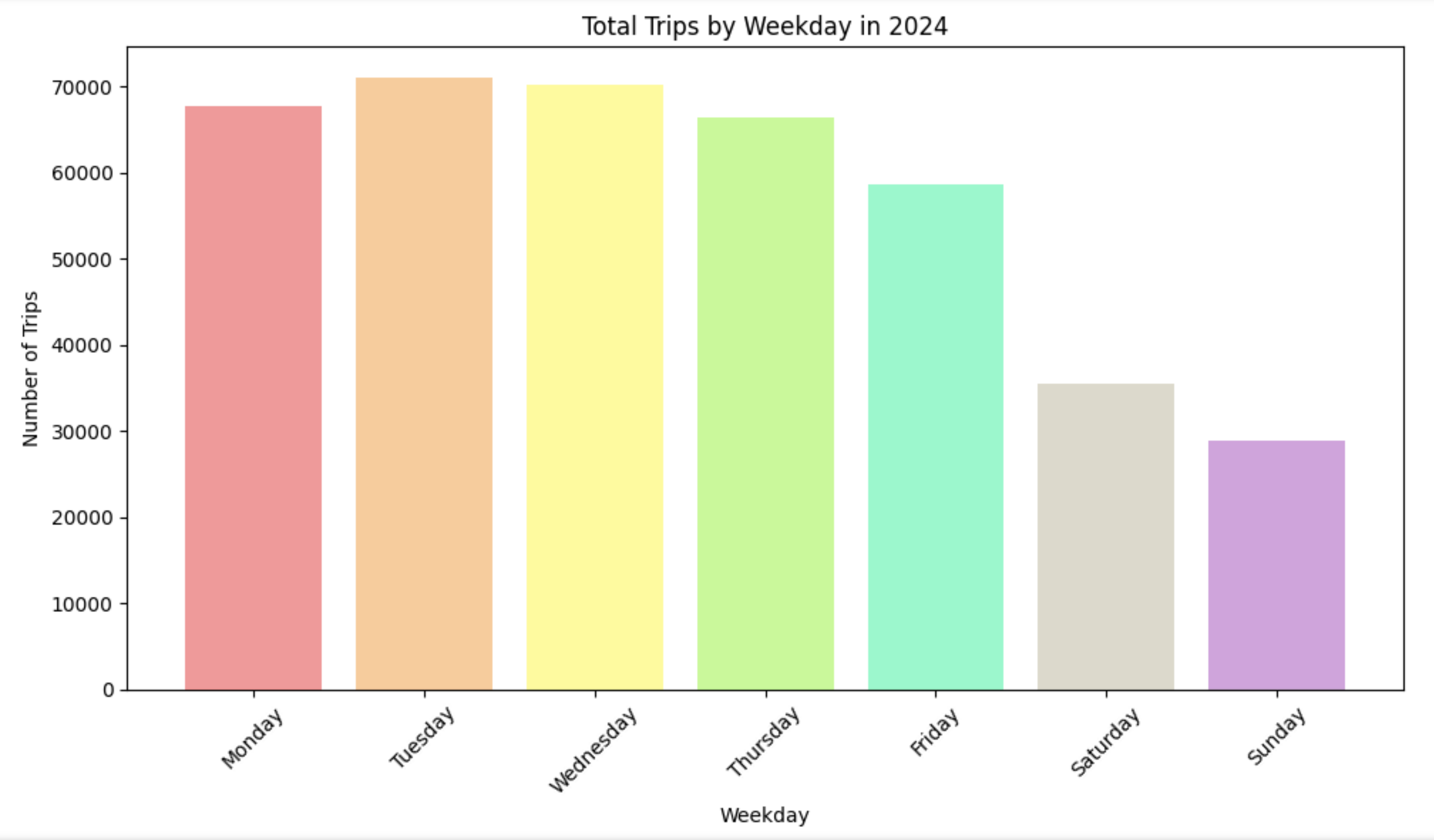
Station Hubs 2024

Least Popular Station Hubs

Folke Bernadottesvei 38
Ortun Svømmehall
Vestlund Borettslag
Dag Hammarskjöldsvei 73
Lynghaug Borettslag
Ortustranden
Sælen
Lynghaugparken
Sælemyr busstopp
Holtet Borettslag

Popular Ride Times (2024) ➔

Peak Hours
7
8
14
15
16
17

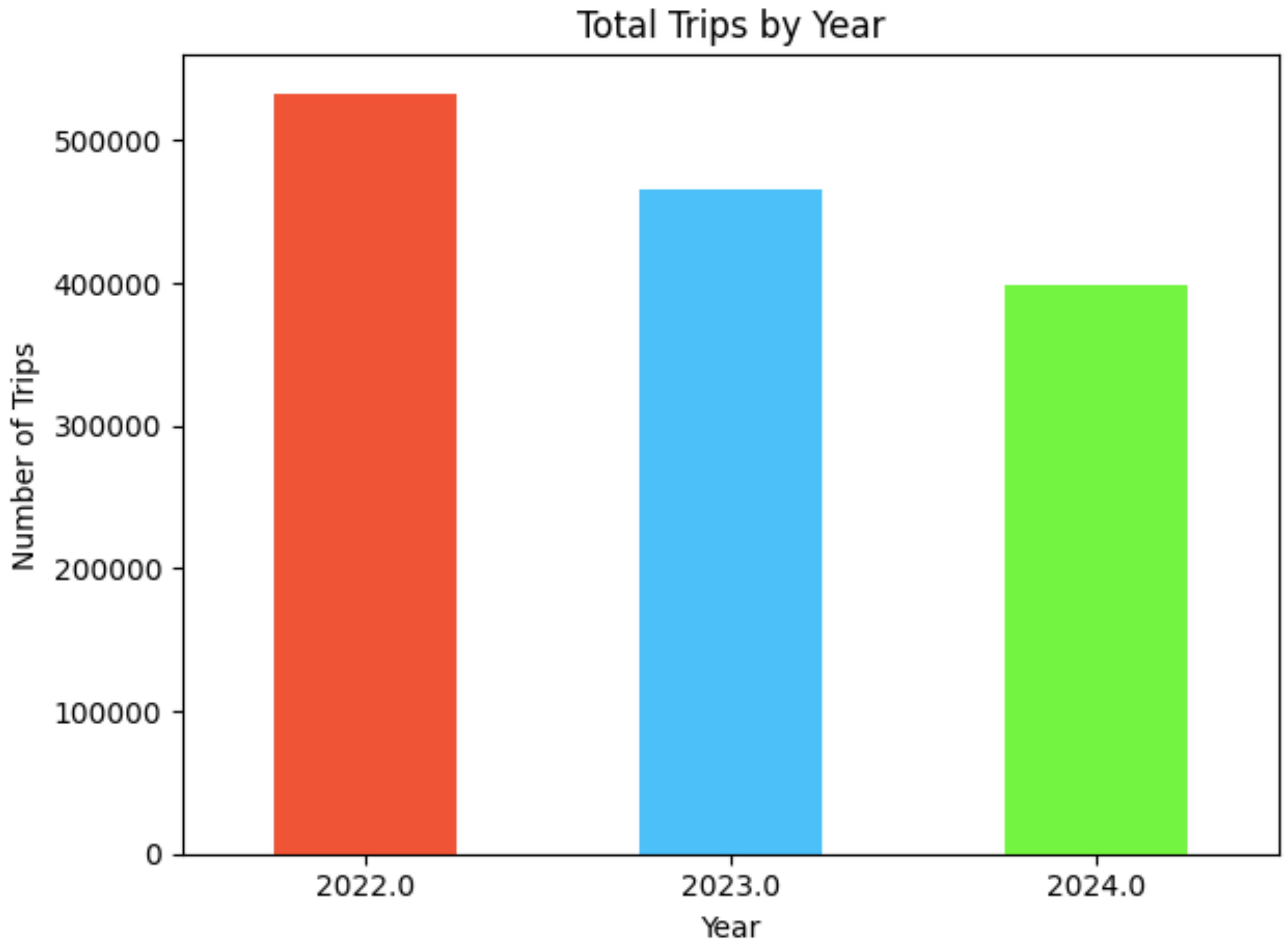


Total Trips by Weekday in 2024
Number of Trips
70000
60000
50000
40000
30000
20000
10000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
Weekday

2022:
533,073

2023:
465,033

2024:
398,486



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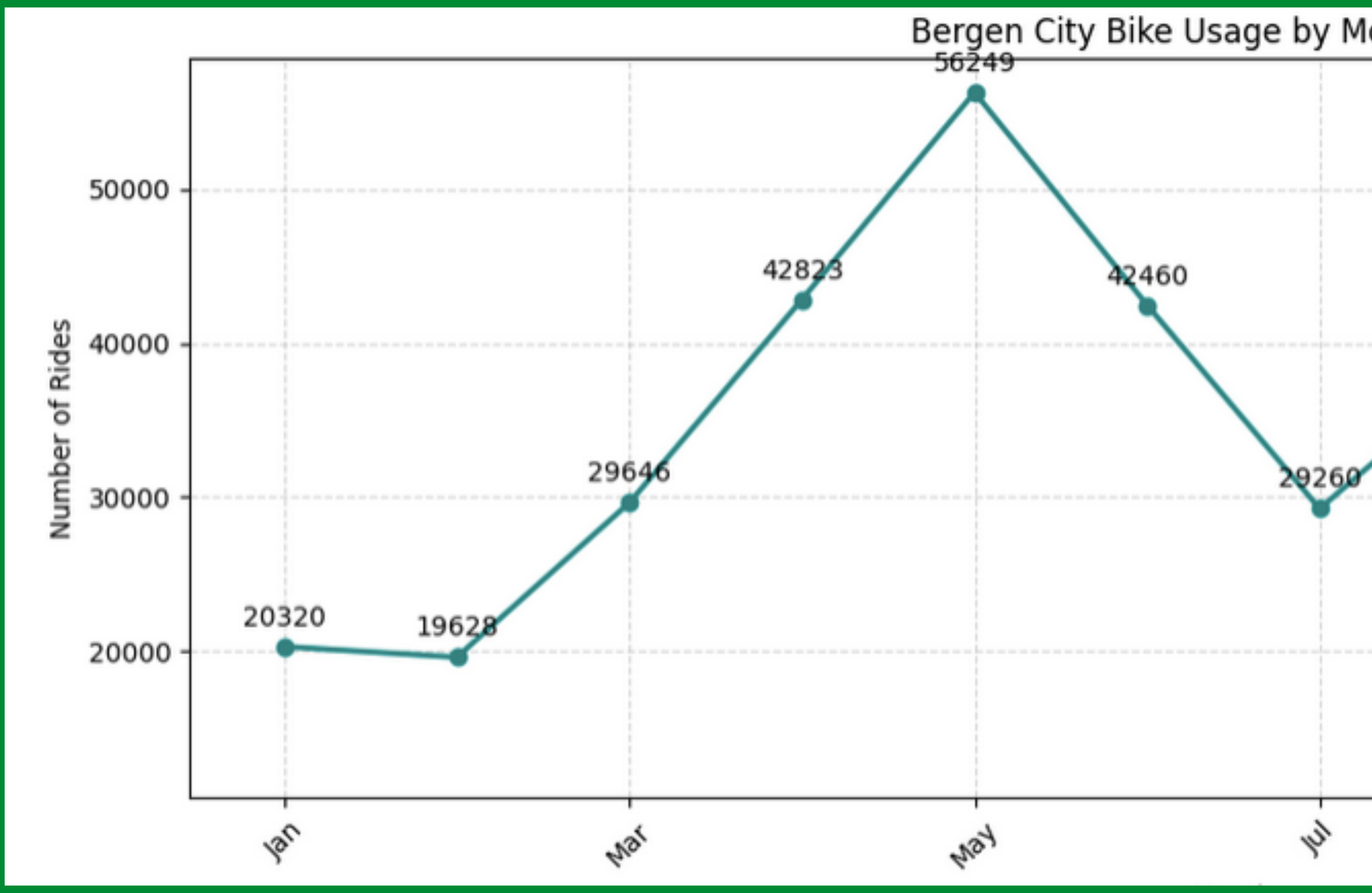
Rides by Month (2024)

Months with th

- May
- Septembe
- April
- June



Future Trends

(12 Month Forecast with Prophet M
Implementation Overview)

Using the Prophet time series model, I forecasted monthly ride volumes through 2025.
holiday effects, and a custom regressor for weekday density — which helps account fo
demand.

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01 Ran grid search across 60 parameter combinations

03 Selected best parameters based on lowest MAE

Model Accuracy

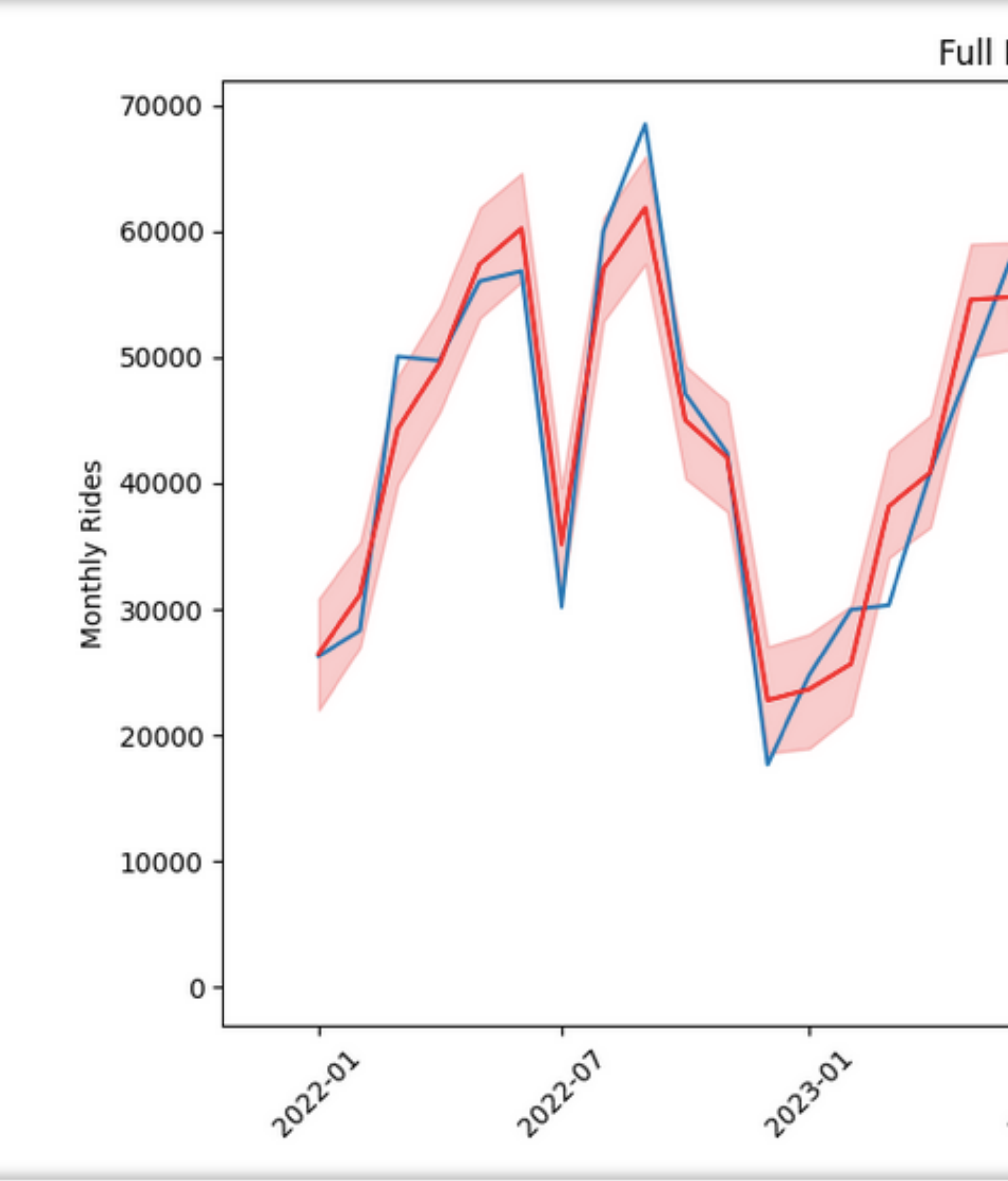
MAPE - 7.3%

- predictions are, on average, within 7.3% of actual observed values.

Predicted rides for 2025:

294,825

(12 Month Foreca
Results)



Total rides are expecte

Database Desig

Use Case Example: Data Analysis - Popular Routes

1. The data analyst logs into the database
2. The data analyst queries the Trip table for start_station_name, end_station_name, start_time, end_time, duration.
3. The data analyst groups the data by start_station_name and end_station_name and counts the number of trips per route.
4. The system returns the routes by total trip count.

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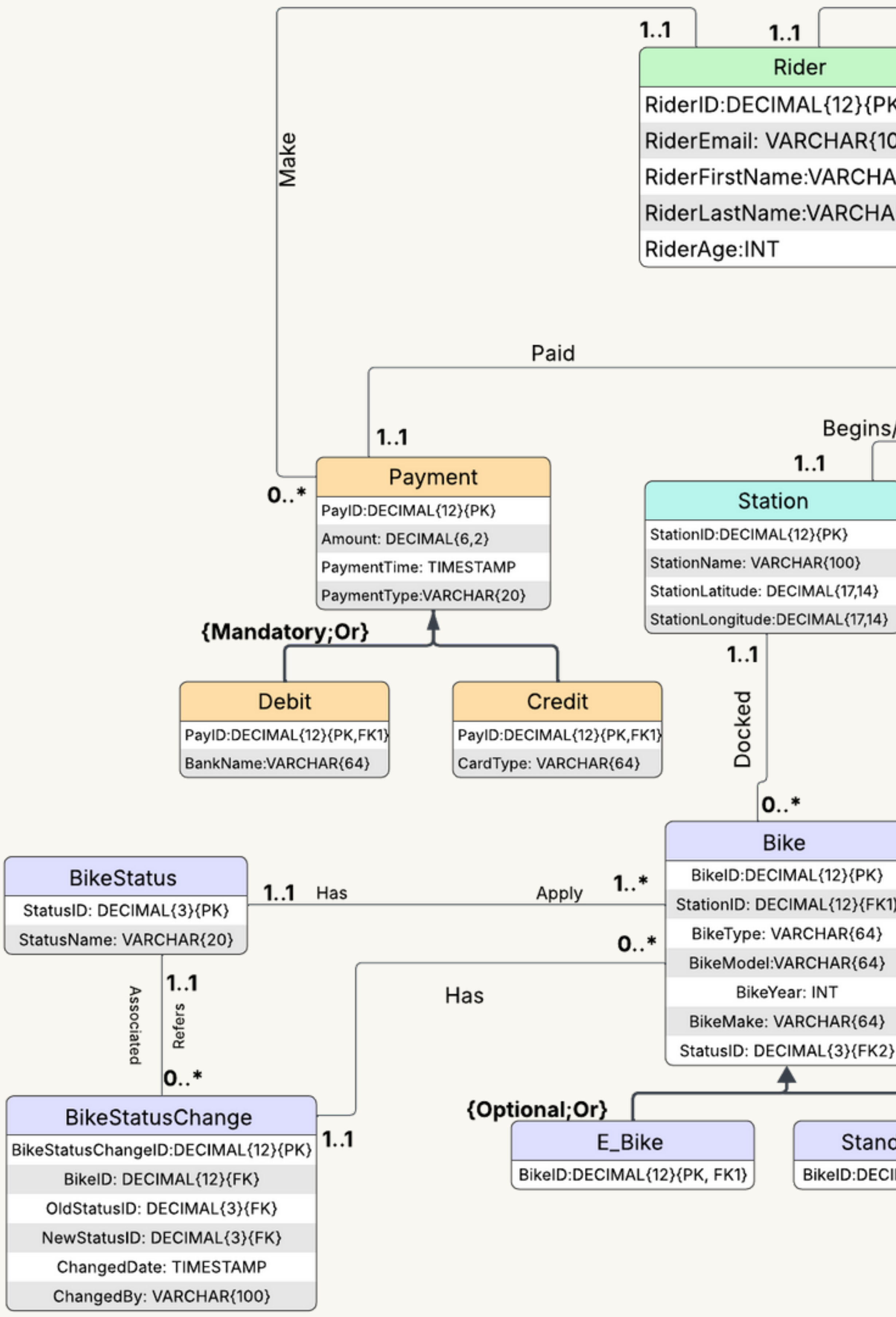

(DBMS Physical ERD)

(Recommendations)

Optimize maintenance scheduling and resource allocat

- Schedule preventive maintenance and upgrades in De February.
- Increase staffing and bike availability in April–June
- Scale back operations in December–February to reduc capacity.

Enhance Data Collection & Integration

- database can be leveraged to enrich forecasting and o decisions.
- Integrate weather, station usage, and rider demograph deeper insights.